

AERIAL OBJECT CLASSIFICATION

PROBLEM STATEMENT:

Bird and drone detection has become an important problem in the field of computer vision and aerial surveillance. With the increasing use of drones in various applications such as photography, delivery systems, agriculture, and security monitoring, it has become necessary to accurately distinguish drones from birds in aerial images. Since birds and drones may appear visually similar in the sky, manual identification can be difficult and time-consuming.

This project focuses on developing a deep learning-based image classification system capable of identifying whether an aerial object is a bird or a drone. The system uses Convolutional Neural Networks (CNN) and transfer learning techniques to perform binary image classification. Two models were implemented and compared in this project: a Custom CNN model and an EfficientNetB0 transfer learning model.

The dataset used in this project consists of labeled bird and drone images. Image preprocessing and augmentation techniques were applied to improve model generalization and performance. The trained models were evaluated using performance metrics such as accuracy, precision, recall, and F1-score.

In addition, a user-friendly web application was developed using Streamlit to allow users to upload images and obtain real-time predictions. The goal of this project is to build an efficient and accurate aerial object classification system that can assist in surveillance, wildlife monitoring, and drone detection applications.

DATASET :

The dataset used in this project consists of aerial images belonging to two classes: Bird and Drone. The images were collected and organized into separate folders for training, validation, and testing purposes.

The dataset was prepared for binary image classification, where

- Class 0 represents Bird
- Class 1 represents Drone

To improve model performance and reduce overfitting, image preprocessing and augmentation techniques were applied. The images were resized to a fixed input size of 224×224 pixels before training.

Data augmentation techniques such as rotation, zooming, shifting, and horizontal flipping were used to increase dataset diversity and improve model generalization.

The dataset was divided into:

- Training set
- Validation set
- Testing set

This division helps evaluate the model's ability to generalize to unseen images.

Models used:

Custom CNN Model

A Custom Convolutional Neural Network (CNN) was developed for binary image classification. The CNN model consists of multiple convolutional layers, max-pooling layers, dropout layers, and dense layers. ReLU activation was used in hidden layers, while the softmax activation function is used in the output layer for binary classification.

The CNN model is trained using the Adam optimizer and binary cross-entropy loss function.

EfficientNetB0 Transfer Learning Model

Transfer learning was implemented using the EfficientNet pretrained model. The pretrained layers were used as feature extractors, and custom classification layers were added on top of the base model.

Transfer learning helps improve performance by utilizing knowledge learned from large-scale datasets. The EfficientNetB0 model achieved better classification accuracy compared to the Custom CNN model.

Evaluation metrics:

The trained models were evaluated using the following metrics:

- Accuracy
- Precision

- Recall
- F1 Score

These metrics were used to compare the performance of the Custom CNN and EfficientNetB0 models.

Result and analysis:

The performance of the Custom CNN model and the EfficientNetB0 transfer learning model was evaluated using multiple evaluation metrics including accuracy, precision, recall, and F1-score.

The Custom CNN model achieved moderate classification performance, while the EfficientNetB0 model significantly improved the prediction accuracy due to transfer learning and pretrained feature extraction capabilities.

The comparison of both models is shown below:

Model	Accuracy	Precision	Recall	F1 Score
Custom CNN	80.46%	80.46%	80.46%	80.21%
EfficientNetB0	97.67%	97.67%	97.67%	97.67%

The EfficientNetB0 model outperformed the Custom CNN model in all evaluation metrics. The higher accuracy demonstrates that transfer learning is highly effective for aerial object classification tasks.

The confusion matrix and prediction results indicate that the EfficientNetB0 model was able to distinguish bird and drone images with high reliability and fewer misclassifications.

The training process also showed that the EfficientNetB0 model converged faster and achieved better generalization performance on unseen test data.

	Model	Accuracy	Precision	Recall	F1 Score	Training Time (min)
0	Custom CNN	0.8047	0.8047	0.8047	0.8021	64.1848
1	EfficientNetB0	0.9767	0.9767	0.9767	0.9767	69.1957

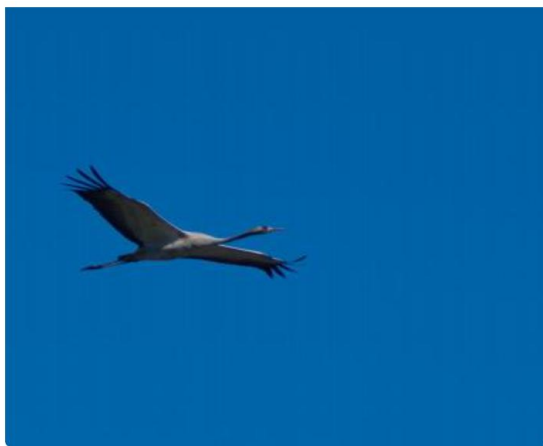
Predicted sample images:



Uploaded Image

Prediction: drone

Confidence: 99.18%



Uploaded Image

Prediction: bird

Confidence: 99.64%



Conclusion:

This project successfully developed a deep learning-based aerial object classification system capable of distinguishing between birds and drones. Two deep learning models, namely a Custom CNN model and an EfficientNetB0 transfer learning model, were implemented and evaluated using multiple performance metrics such as accuracy, precision, recall, and F1-score.

The experimental results demonstrated that the EfficientNetB0 model significantly outperformed the Custom CNN model, achieving higher classification accuracy and better generalization performance on unseen test images. The use of transfer learning improved feature extraction and reduced classification errors.

A user-friendly Streamlit web application was also developed to provide real-time image upload and prediction functionality. The system can effectively classify aerial images as birds or drones with high confidence.

Overall, this project highlights the effectiveness of deep learning and transfer learning techniques in solving real-world aerial object classification problems and provides a strong foundation for future research in drone surveillance and intelligent monitoring systems.